
MINTS: What Evidence Supports a Genomic Attention Head as a Motif Detector?

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Abstract

What evidence is required before calling a genomic-transformer attention head a biological motif detector? We distinguish label decodability, motif-local association, and intervention effects on a specified output. MINTS tests this distinction using frozen DNABERT-2 representations, token-aligned CTCF motif scores, and promoter perturbations. Sequence-level bootstrap intervals accompany the predictive results. Simple k-mer classifiers have higher point AUROC than the frozen readout on both promoter tasks, and composition matching reduces promoter readout performance. No head passed the historical motif-local QK and attention-enrichment screens, whose approximation limits biological interpretation. A separate validated native-attention assay finds positive associations in 58 heads across 5,220 matched motif-present/absent peak pairs, after 144-head correction. These conditional associations do not establish motif detection or native CTCF causality. Held-out promoter patching measures heterogeneous effects on a trained probe, with sign-changing clean-corrupted denominators. The study provides an evidence standard and an auditable example of why predictive accuracy, attention association, and probe restoration support different claims.

1 Introduction

A motif is a recurring short DNA pattern associated with a biological process. A genomic transformer can classify DNA containing such a pattern without assigning that computation to one attention head. A linear probe may recover the label from frozen hidden states, while attention maps may emphasize nearby bases. Neither observation shows that the head detects the motif or that the model uses its output for a biological prediction.

We ask what evidence would justify that stronger component-level claim. The distinction between the query–key computation that directs attention and the output–value computation that writes information motivates separate tests of localization and downstream effect [1]. Activation patching contributes intervention evidence only after the output, corruption, and component being exchanged have been specified [2]. Our contribution is an explicit standard for reporting these different forms of evidence, together with an audit of a DNABERT-2 case study [3].

CTCF is a protein that binds DNA and contributes to chromosome organization. A CTCF peak is a genomic interval enriched in a binding assay; a peak need not contain a high-scoring match to one motif matrix. We use the curated JASPAR MA0139.1 matrix to define a testable motif-local hypothesis. Promoters regulate transcription near gene starts, and a TATA box is one promoter-associated A/T-rich pattern. Splice donor and acceptor signals mark boundaries removed when RNA is processed. These auxiliary classification tasks test representation decodability, rather than substituting for a CTCF-specific causal experiment.

The evidence standard is a reporting discipline, not an exhaustive definition of every possible motif detector. A distributed detector, a detector active at particular queries, or a detector expressed through nonlinear interactions can fail our screens. We distinguish failure to pass a specified assay from absence of a biological mechanism.

Table 1: Evidence required for increasingly specific claims. Cutoffs describe historical screens, not established biological boundaries.

Claim	Required observation and scope
Label decodability	A held-out classifier over frozen hidden states, compared with composition and sequence baselines.
Motif-local association	QK/motif association and enrichment over explicit backgrounds; historical screens use $r \geq 0.5$ and an enrichment ratio ≥ 2 .
Causal contribution	An intervention changes a specified output on held-out sequences, with uncertainty and control interventions. Probe effects concern the probe.
Biological motif detector	Aligned localization and causal evidence for the same motif, with motif-present and motif-absent controls and a validated native target.

2 Methods

Evidence and output targets. Table 1 separates the claims. Historical screening cutoffs are heuristic. We found no timestamped registration evidence before examining results; the original threshold-choice date is unknown. The threshold sweep is the principal descriptive robustness analysis, and does not replace sequence-level inference or replication.

Predictive comparisons. The encoder is zhihan1996/DNABERT-2-117M, which uses variable-length byte-pair tokens. We mean-pool non-padding hidden states, including special tokens, at zero-based encoder layer 11, then train a standardized, class-balanced logistic classifier. The earlier labels “Seq-head” and “Readout” described this same classifier, so we report it once as a frozen readout. No encoder fine-tuning is performed.

The composition baseline uses GC fraction, GC skew, and sequence length. GC denotes the fraction of valid bases that are G or C. The sequence baseline uses TF-IDF counts of overlapping character 3–6-mers and balanced logistic regression. This range covers short local patterns at manageable feature cost; it was a heuristic configuration, not an optimum chosen using an independent validation set. TF-IDF is fit on training sequences only, with minimum document frequency 2 and at most 50,000 features. Solver settings, seeds, and predictions are saved with the regenerated artifacts.

Historical nearest-GC matching without a distance limit merely reorders both balanced promoter test sets. We replace it with matching without replacement within 0.02 GC fraction, introduced heuristically during this revision, and report the retained population. This changes the estimand to sequences with composition overlap, and does not eliminate other confounding features.

Uncertainty and selection. Predictive intervals are percentile intervals from 1,000 resamples of test sequences, seed 1729, conditional on a fixed trained classifier. They exclude training-set and checkpoint variability. Undefined single-class draws are omitted for AUROC and AUPRC. Sequences, rather than their individual tokens, are resampled. Exploratory head-level intervals are marginal; choosing the largest effect among 144 heads does not make its interval evidence of discovery. Confirmatory head searches require sequence-level null inference with correction across all tested heads, such as Holm family-wise correction. Historical token-level Pearson p-values are not used for significance claims here.

Nucleotide-to-token alignment. A position weight matrix gives a nucleotide preference at each motif position. We normalize MA0139.1 counts with pseudocount 0.5 and scan both orientations with log-odds scores. The historical support threshold is $s_{\min} + 0.8(s_{\max} - s_{\min})$. A hit spans $[a, b]$ and a token spans $[u, v]$. It belongs to motif support when $\max(0, \min(v, b) - \max(u, a)) \geq 1$. Zero-width special tokens remain aligned to hidden states but receive no support. Each token’s score is the maximum over overlapping scored motif windows.

Historical CTCF assay. For every head, the assay averages bilinear QK scores over all non-padding queries, including special tokens, and correlates resulting key scores with finite motif scores pooled across sequences. Its attention map is reconstructed from QK weight factors. Equivalence to native attention, including projection biases, positional terms, and normalization, has not been established. Enrichment therefore describes the implemented reconstruction.

Table 2: Exact stored class counts. Membership, coordinates, sequence hashes, and orientation-canonical hashes are saved as JSONL.

Task	Train positive	Train negative	Test positive	Test negative
TATA promoter	2,531	2,531	106	106
Other promoter	14,980	15,020	686	686
Splice donor	15,084	14,916	1,480	1,520
Splice acceptor	15,031	14,969	1,475	1,525

Background positions are the nearest token indices outside each individual motif-hit interval, matching token count. They are not matched for GC, nucleotide span, or accessibility, and can overlap other hits or special tokens. Enrichment is the ratio of aggregate attention mass per motif-hit token occurrence to mass per background occurrence, summing over queries. Longer sequences and overlapping hits affect this historical weighting. This ratio, sometimes denoted ρ , is not Spearman correlation and can exceed one.

Native-attention genomic controls. We separately match motif-absent peaks without replacement to motif-present peaks on chromosome, GC fraction within 0.02, and nucleotide length within 5%. The first connected motif-support span in the present sequence is projected to the same relative position in the absent sequence. These are PWM-absent binding peaks, not unbound genomic regions; accessibility and other sequence features remain possible confounders.

For checkpoint revision 7bce263b15377fc15361f52cfab88f8b586abda0, attention reconstruction includes projection biases and the native positional bias and mask. We check reconstructed attention times native values against every native head-context output. Queries and target keys span nucleotides and exclude special tokens; softmax still includes native special-token keys. The score averages attention over real queries and target keys, multiplied by the number of real keys to normalize uniform attention density. Each matched sequence pair is one unit. We use 1,000 pair-bootstrap draws for marginal intervals and 9,999 paired sign flips for two-sided tests, with Holm correction over 144 heads. Conditional pair exchangeability is an observational assumption, and within-chromosome dependence is not modeled. These new matching and inference settings were chosen during revision.

Patching procedure and interpretation. We replace the first TATA-like pattern matching TATA[AT]A with a deterministic G/C-rich sequence of the same nucleotide length. This changes composition as well as motif identity. We retain positive held-out examples whose clean and corrupted token counts agree; equal counts alone do not establish identical nucleotide boundaries. Clean head activations are inserted into the corrupted run at selected positions. The scalar is the trained standardized logistic probe’s decision function on the mean-pooled layer-11 residual vector, not a pretrained native output.

$$\text{PM} = \frac{f(x_{\text{patched}}) - f(x_{\text{corrupted}})}{f(x_{\text{clean}}) - f(x_{\text{corrupted}})}. \quad (1)$$

Near-zero denominators, absolute effects, and denominator signs must be examined alongside PM. Values above one indicate overshoot and can reflect off-manifold combinations or denominator sensitivity. Clean/corrupted scores, pair effects, IDs, medians, and bootstrap intervals are saved. No native CTCF activation-patching result is claimed because a validated native CTCF output is not defined.

3 Datasets and audit scope

We use locally saved partitions of InstaDeepAI/nucleotide_transformer_downstream_tasks_revised, filtering the task field and retaining sequence, coordinate name, and binary label. Saved data have train and test partitions, with no independent validation partition. Test examples are on chromosomes 20 and 21; training examples are on chromosomes 1–19, X, and Y. Table 2 gives exact class counts. The upstream revision and every source label’s provenance were not pinned in the original run; local membership hashes establish analyzed data, not an independently reconstructed upstream release.

Table 3: Held-out AUROC with 95% sequence-bootstrap intervals, generated from saved per-sequence predictions.

Task		GC	3–6-mer	Frozen readout
TATA promoter	0.8955 [0.8496, 0.9337]	0.9297 [0.8947, 0.9598]	0.9137 [0.8751, 0.9475]	
Other promoter	0.9088 [0.8937, 0.9236]	0.9406 [0.9292, 0.9517]	0.9383 [0.9253, 0.9499]	
Splice donor	0.6560 [0.6375, 0.6750]	0.8185 [0.8046, 0.8320]	0.8954 [0.8839, 0.9060]	
Splice acceptor	0.6361 [0.6156, 0.6547]	0.7956 [0.7792, 0.8113]	0.8847 [0.8723, 0.8959]	

Table 4: Readout AUROC and 95% sequence-bootstrap intervals. Full, historical, and caliper populations differ; retained counts and composition diagnostics accompany the predictions.

Task		Full	Historical matching	GC caliper
TATA promoter	0.9137 [0.8751, 0.9475]	0.9137 [0.8737, 0.9477]	0.7787 [0.6662, 0.8681]	
Other promoter	0.9383 [0.9253, 0.9499]	0.9383 [0.9253, 0.9496]	0.7254 [0.6797, 0.7711]	
Splice donor	0.8954 [0.8839, 0.9060]	0.8944 [0.8826, 0.9054]	0.8600 [0.8449, 0.8751]	
Splice acceptor	0.8847 [0.8723, 0.8959]	0.8838 [0.8717, 0.8960]	0.8607 [0.8456, 0.8750]	

Checks find no shared chromosomes, overlapping windows, identical sequences, or reverse-complement duplicates across stored train/test partitions. An exhaustive equal-length Hamming screen finds no cross-partition pair with at most two substitutions in either orientation. This does not cover indels, shifted windows, or longer-range homology. No reverse-complement augmentation is introduced. Chromosome-held-out evaluation cannot rule out reference-genome overlap in pretraining.

CTCF sequences come from ENCODE experiment ENCSR000DKV, peak file ENCF827JRI, and the UCSC hg38 reference. BED intervals are extracted without flanks; the prepared file contains 51,249 intervals. Peaks are assay observations rather than proof of a particular motif or a model’s binding prediction. Local source hashes are audited; unpinned model revisions limit exact checkpoint reconstruction.

4 Results

Prediction is context, not a mechanism. Table 3 shows higher point AUROC for the k-mer classifier than the DNABERT-2 readout on both promoter tasks; this is not a claim of statistically established superiority. The readout has higher point AUROC on splice tasks. Strong promoter composition baselines make a motif-specific explanation especially uncertain.

Composition matching changes the promoter result. Table 4 retains the historical match for auditability and compares the repaired caliper control. Historical and full promoter AUROC legitimately agree for identical memberships, but this provides no composition control. Historical splice subsets are smaller and their AUROCs differ from full scores. The caliper reduces promoter readout performance; the GC baseline on the retained population approaches chance. This supports a narrower promoter interpretation and does not establish causal motif use.

CTCF screen failure is scoped to this assay. The token-score file has 1,946,372 rows, including 1,843,874 finite scores and 256,918 unique motif-support tokens. Summing participation separately over overlapping hits gives 281,915 motif-hit token occurrences, the quantity used by historical enrichment. There are 5,344 peaks without a threshold-passing PWM hit. These count different filtering and aggregation stages.

The maximum historical QK correlation is 0.3004 and reconstructed enrichment ratio is 1.3130. No head passed the motif-local QK and attention-enrichment screens. Table 5 gives the joint sweep from saved head tables. This does not establish absence of a CTCF detector. We withdraw inferential interpretations of old token-shuffle and permuted-head nulls, which do not provide sequence-level evidence for localization.

Native attention associates with motif-present peaks. The separate genomic-control assay retains 5,220 pairs, with no invalid pairs. Reconstructed and native head contexts agree to the measured

Table 5: Heads passing both descriptive cutoffs in the historical reconstructed-attention assay. No significance or confidence interval is assigned to a count of screened heads.

QK r cutoff	Ratio 1.1	Ratio 1.25	Ratio 1.5	Ratio 2.0
0.1	2	1	0	0
0.2	0	0	0	0
0.3	0	0	0	0
0.4	0	0	0	0
0.5	0	0	0	0

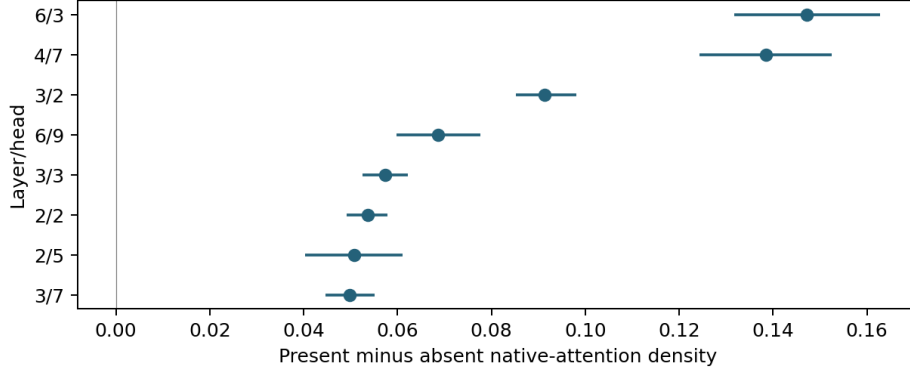


Figure 1: Native-attention density contrasts for the eight largest observed means. Intervals bootstrap matched sequence pairs and are marginal; tests correct across all 144 heads.

precision (maximum absolute error zero). There are 58 heads with positive contrasts and Holm-adjusted $p < 0.05$. The largest mean contrast, layer/head 6/3, is 0.1471 (marginal 95% interval [0.1317, 0.1628]; Holm $p = 0.0144$). Figure 1 shows the largest observed contrasts. These conditional associations qualify the historical negative result, but do not supply validated motif-local QK evidence or a native causal output. Selection of displayed maxima is exploratory.

Patching is a held-out probe experiment. The earlier 327 TATA pairs contain 317 training and 10 test examples, with unique pair IDs; pooling partitions explains the apparent excess over the test-set size. That pooled effect is withdrawn as held-out causal evidence. Revised test-only results retain the 10 eligible examples. Table 6 reports the largest mean effects with medians and marginal intervals. Figure 2 shows individual pairs and their denominators. Sign changes in the reference difference and variation across pairs limit interpretation of the mean ratio. The intervention changes a trained readout and cannot certify a biological detector in the pretrained model.

Table 6: Exploratory test-only TATA probe patching, ranked by mean PM. Intervals bootstrap pairs and are marginal, without correction for choosing maxima across 144 heads.

Layer/head	Mean PM	Median PM	Mean 95% CI	Pairs
2/7	0.3499	0.0298	[-0.0264, 0.9909]	10
11/4	0.3034	0.0347	[-0.1203, 0.9401]	10
3/10	0.2935	0.0381	[0.0173, 0.6532]	10

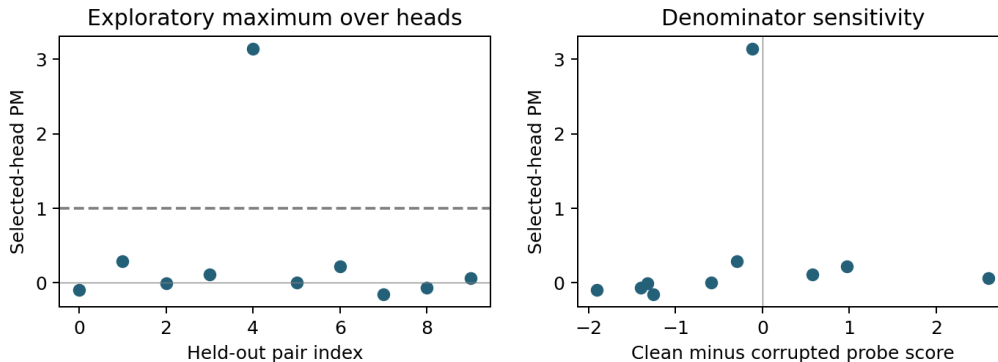


Figure 2: Pair PM for the head with the largest observed mean, and its clean-corrupted score difference. Selection is exploratory.

5 Interpretation and limitations

Evidence supports frozen decodability, failure of historical CTCF screens, and native-attention association under matched genomic controls, without identifying native CTCF causality. Sequence-level QK inference and a validated causal target remain unresolved. The corrected native-attention analysis uses a different aggregation and control population from the historical screens; the two assays are not interchangeable. Small selected patching populations and composition-changing corruptions prevent broad causal conclusions. Future patching must establish nucleotide alignment and contrast motif edits with matched non-motif edits.

The affine QK observation is conditional: if logits for a fixed query equal $a + \gamma s_j$ with $\gamma > 0$, softmax preserves score ordering within that sequence. This assumption is not shown empirically. Within-sequence ranking does not imply a cross-sequence motif-present/absent threshold or causal use; we make no stronger layer-1 theorem claim.

The Nucleotide Transformer comparison is excluded because a pinned checkpoint, validated token alignment, and native-attention equivalence were not established. Its stored enrichment ratio is not Spearman correlation. No tokenizer-family ranking or cross-model biological conclusion is supported. Auxiliary OV and sparse-feature artifacts are historical; weak alignment does not show that an alternate circuit is absent.

Intervals concern fixed classifiers and do not resolve labeling errors, pretraining contamination, within-chromosome dependence, or training-seed variability. Local membership is documented, but upstream pinning and independent label reconstruction remain incomplete. Tables were regenerated from frozen caches. After successful installation and ingestion, the uncapped clean pipeline failed during an ENCODE download because the disk filled; it is not a completed reproduction.

6 Availability, licenses, and declarations

Availability. Code, configuration, machine-readable audits, predictions, generated tables, commands, hashes, and failures accompany the revision. Large genomic/model artifacts are obtained from upstream sources under their own terms. The submitted state and revised artifacts are distinguished.

Table 7: Resource terms. Unverified third-party terms are not replaced by the MINTS code license.

Resource	License and verification scope
MINTS original code	MIT; excludes third-party data, models, and venue style files.
DNABERT-2	Upstream code: Apache-2.0. A separate checkpoint license is not established by its card.
Nucleotide Transformer v2 100M Revised downstream dataset	Model card: CC-BY-NC-SA-4.0; comparison excluded. Revision-specific license not independently established; consult original sources.
ENCODE / UCSC hg38 JASPAR MA0139.1	Publicly accessible; source-specific terms apply. Public matrix; upstream terms apply.

This revision has not been published to the anonymized repository, so no existing external link is asserted to contain it.

Ethics. This analysis uses existing reference sequences and assay intervals. It does not recruit participants or support clinical decisions. No new ethics approval is claimed. Biological claims are limited to the implemented measurements.

Contributions. This is a sole-author study. The author directs the research and is responsible for final scientific claims; human verification of this assisted revision remains required.

Funding and conflicts. The author reports no funding and no conflicts of interest.

AI use. Codex assisted this revision with code/artifact inspection, implementation, command execution, and text drafting. Generated interpretations require author review. Specific earlier model versions and prior human verification are not independently documented, so no such attestation is made.

7 Conclusion

Calling a genomic head a biological motif detector requires connecting motif-local measurements to interventions on a validated output for the same motif. Frozen prediction and probe restoration support narrower claims. No head passed the historical motif-local QK and attention-enrichment screens, while a separate native-attention assay finds conditional genomic associations. Native CTCF causality and sequence-level QK evidence remain unresolved.

References

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